

Document	System Build Plan
Project	Luxury Retail Employee Flash Sale Operations System
Developer	Nandani Raj
Tools	Power Apps · Power Automate · SharePoint · Power BI · SQL
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Version	1.0

1. Architecture Overview

The system follows a four-layer architecture with clear separation of responsibilities. Each layer has a single defined role, which makes the system easier to maintain, extend, and troubleshoot independently.

Layer	Technology	Responsibility
Presentation	Power Apps Canvas App	All user interaction — role-based screens, input forms, validation logic, navigation
Process	Power Automate	All automated business logic — approval routing, notifications, reminders, cleanup triggers
Data	SharePoint Online	All persistent data — 8 structured lists as the system of record
Reporting	Power BI	All operational reporting — refreshable dashboard, DAX measures, star schema model

All four layers connect through SharePoint: the app reads and writes to SharePoint lists, Power Automate flows are triggered by SharePoint item events, and Power BI connects to SharePoint as its data source. This means SharePoint is the integration point — getting the list structure right before building anything else is the most critical step in the build sequence.

2. SharePoint Backend Summary

The backend consists of eight structured SharePoint Online lists. All lists are linked through a shared SaleID text field that acts as a foreign key across the system. Full column specifications are in docs/SharePoint_Backend_Schema.pdf.

List	Purpose	Key Relationships
Sale Master	Authoritative record for every flash sale event	Primary — SaleID is the key referenced by all other lists
Employee Access Requests	All access requests submitted by employees	SaleID → Sale Master; RequestID → Approval Log
Approval Log	Audit record of every approval decision	RequestID → Access Requests; SaleID → Sale Master
Issue Tracker	All operational issues logged during a sale	SaleID → Sale Master

List	Purpose	Key Relationships
Notification Log	Record of all automated emails sent by flows	SaleID → Sale Master
Change History	Audit log of all status changes and key updates	RecordID maps to Sale Master or Access Requests
Sale Readiness Checklist	Pre-launch checklist items per sale	SaleID → Sale Master
User Role Reference	Role assignments for all app users	UserEmail — looked up via Office() on app load

3. Power Apps Module Summary

The Canvas App has ten screens. Navigation is controlled by a varCurrentUser variable set on App.OnStart by looking up the logged-in user's email in the User Role Reference list. What each user sees depends on their assigned role.

Note: All Power Fx formulas in the build specification are draft implementation patterns. They must be tested against final SharePoint internal column names and control names before deployment.

Screen	Primary User	Key Function
Home / Dashboard	All roles	Lists active and upcoming sales; routes to role-appropriate screens
Sale Setup	Operations Admin	Create new sale record — all required fields with validation
Sale Details	All roles	View sale info, dates, discount, product categories, participant count
Access Request	Employee	Submit access request; eligibility validation before submit
Access Management	Operations Admin	View, filter, and action the access request queue
Issue Tracker	All roles	Log and view operational issues; priority classification
Issue Details	Admin / Support	Update issue status, add resolution notes, escalate
Sale Readiness Checklist	Operations Admin	Work through pre-launch checklist; Authorize Launch button
Notification / Status	Employee	View current request status and access confirmation
Admin Control Panel	Operations Admin	Summary dashboard with navigation to all admin functions

Security note: App-level role visibility is not the same as true data security. A production deployment must combine app-level navigation control with Power Apps sharing configuration, SharePoint site permissions, and Microsoft 365 group membership.

4. Power Automate Workflow Summary

Seven cloud flows handle all automated business logic. All flows use the SharePoint and Outlook connectors. Flow steps are triggered by item creation or modification events in SharePoint lists.

#	Flow Name	Trigger	Primary Action
1	Access Request Approval	New request created with status Pending	Route approval email to manager; update status; notify employee

#	Flow Name	Trigger	Primary Action
2	Sale Setup Approval	SaleStatus changes to 'Open for Requests'	Notify ops team; confirm setup completeness
3	Launch Reminder	Scheduled: 24 hours before SaleStartDate	Email approved employees with sale access confirmation
4	High-Priority Issue Alert	New issue with Priority = 'High' created	Immediate alert email to ops admin and IT support
5	Status Notification	RequestStatus changes on any request	Email employee with updated status and reason
6	Post-Sale Cleanup Reminder	SaleStatus changes to 'Closing'	Email admin with list of employees needing access revocation
7	Daily Active Sale Summary	Scheduled: daily at 8:00 AM during active sales	Email admin with open issues, pending requests, readiness summary

5. SQL Validation Scripts Summary

Nine T-SQL validation scripts provide data quality checks, SLA monitoring, and operational reporting support. Scripts are designed to run against a SQL Server database that mirrors the SharePoint list structure.

#	Script Name	Purpose
01	Duplicate Active Access Records	Identifies employees with more than one active access record for the same sale
02	All Pending Requests — SLA Status Check	Returns all pending requests with SLA label: Within SLA / Approaching Breach / SLA BREACHED
03	Expired Access Still Active	Flags approved access records where the sale end date has passed
04	Open High-Priority Issues	Returns all unresolved High priority issues with time elapsed since logging
05	Incomplete Sale Setup	Validates Draft sales against required fields — flags any missing values
06	Sale Readiness Validation	Calculates readiness score for sales in Readiness Review status
07	Average Issue Resolution Time	Reports average, min, and max resolution hours by sale, category, and priority
08	Access Requests by Status	Summary of requests by sale and status — approval rate, denial rate, pending backlog
09	Issues by Category and Priority	Aggregates issues by category and priority — volume and resolution time trends

Note on Script 02: The SLA check returns ALL pending requests — there is no pre-filter by age. Every pending request is labelled by SLA status: under 24 hours is Within SLA, 24–47 hours is Approaching Breach, 48 hours or more is SLA BREACHED. This ensures the output is useful for triage regardless of how long requests have been pending.

6. Power BI Dashboard Summary

A four-page refreshable dashboard connects to SharePoint Online via the SharePoint list connector. The data model uses a star schema with the Sale Master list as the central table. All other lists relate to Sale Master via SaleID.

Boolean field note: SharePoint Yes/No columns (such as IsComplete in the Readiness Checklist) may import into Power BI as TRUE/FALSE, 'Yes'/'No' text, or 1/0 integers depending on the connector version. Check the actual imported data type in the Power BI model view before writing DAX measures that reference these columns.

Page	Audience	Key Visuals and KPIs
1 — Sale Operations Overview	Operations Admin, Management	KPI cards: Active Sales, Sales Open for Requests, Open Issues, High-Priority Issues. Table: sales by status. Bar chart: issue volume by sale.
2 — Access and Approvals	Admin, Approving Managers	KPI cards: Total Requests, Approval Rate, Pending Requests, Avg Approval Time (hours). Donut: requests by status. Column: requests by department.
3 — Issues and Support	Admin, IT Support	KPI cards: Open Issues, Escalated Issues, Avg Resolution Hours, Resolution Rate %. Stacked bar: issues by category and priority. Line: issue volume over time.
4 — Readiness and Post-Sale	Operations Admin	KPI cards: Readiness Score %, Items Complete, Items Remaining. Table: checklist detail. Table: post-sale access cleanup status.

7. Build Sequence

The phases are ordered to minimise rework. The data structure must be finalised before building the app, the app before automation, and automation before reporting. Documentation is completed after the build is done so it describes what was actually built.

Phase 1 — Foundation

Project charter, system design, SharePoint schema finalisation. All design decisions documented before any build work begins.

Phase 2 — SharePoint Backend

Create all 8 lists with correct columns, choice values, and permissions. Import sample data and verify lookup relationships.

Phase 3 — Power Apps

Build Canvas App connected to SharePoint. Start with Home and Admin Control Panel. Build all 10 screens with Power Fx validation logic. Test role-based navigation.

Phase 4 — Power Automate

Build all 7 flows. Start with Flow 1 (Access Request Approval — most complex). Test each flow end-to-end before moving to the next.

Phase 5 — SQL Scripts

Set up SQL Server with mirrored table schemas. Import sample data. Write, test, and annotate all 9 validation scripts.

Phase 6 — Power BI

Connect Power BI Desktop to SharePoint lists. Build data model. Create DAX measures before building visuals. Build all 4 dashboard pages. Publish and configure refresh.

Phase 7 — Documentation

User guide, admin guide, troubleshooting guide, data dictionary, architecture documentation. Written after build so descriptions reflect what was actually built.

Phase 8 — Portfolio

GitHub README, portfolio case study, screenshots. Produced last — descriptions should reflect the completed system.

8. Testing Checklist

Complete each category in order. Log any failures as issue records in the Issue Tracker to simulate real operational support.

SharePoint Backend

#	Test	Expected Result
SP-1	Create all 8 lists with correct columns, types, and Choice values	Lists visible in site contents with all columns
SP-2	Add 3–5 sample records to Sale Master — verify SaleID uniqueness	No duplicate SaleIDs
SP-3	Confirm column internal names match schema exactly	No binding errors in Power Apps
SP-4	Verify User Role Reference list contains all test accounts	Role lookup works for all test users

Power Apps

#	Test	Expected Result
PA-1	Log in as Employee — verify Home shows only Active and Open for Requests sales	Draft sales not visible
PA-2	Log in as Admin — verify Admin Control Panel and Authorize Launch button visible	Admin-only screens accessible
PA-3	Submit access request as Employee — verify record appears in SharePoint	New record in Access Requests list
PA-4	Submit request for ineligible department — verify validation message	Error shown; no record created
PA-5	Admin completes Readiness Checklist — Authorize Launch button becomes active	Button enabled when all items complete
PA-6	Log in as Read-Only — verify no edit controls visible	View-only experience

Power Automate Flows

#	Test	Expected Result
FL-1	Submit access request — verify Flow 1 triggers and approval email arrives at manager	Email received within 2 minutes
FL-2	Approve request via email — verify RequestStatus updates and employee receives notification	Status updated; notification delivered
FL-3	Create High priority issue — verify Flow 4 triggers alert email to admin	Alert email received immediately
FL-4	Change SaleStatus to Closing — verify Flow 6 triggers cleanup summary email	Cleanup email with correct employee list
FL-5	Check Notification Log — verify all sent notifications are recorded	All flows logging to Notification Log

Power BI Dashboard

#	Test	Expected Result
BI-1	Connect to all 8 SharePoint lists — verify all tables import without errors	All tables visible in data model view
BI-2	Check IsComplete column type — confirm and update DAX accordingly	Readiness Score DAX matches imported type

#	Test	Expected Result
BI-3	Verify Readiness Score = 100% for a fully-completed sale	100% displayed when all items complete
BI-4	Apply SaleName slicer — verify all pages filter correctly	All visuals respond to slicer

9. Screenshots to Capture

Capture these during the build for use as portfolio and GitHub evidence.

What to Capture	Suggested Filename
Power Apps — Home screen (employee view)	app_home_employee.png
Power Apps — Access request with department validation error	app_access_request_validation.png
Power Apps — Admin control panel	app_admin_panel.png
Power Apps — Readiness checklist complete, Authorize Launch active	app_readiness_complete.png
Power Apps — Issue tracker with High priority open issue	app_issue_tracker.png
Power Automate — Flow 1 full canvas view	flow_access_approval.png
Power Automate — Flow run history showing all flows successful	flow_run_history.png
SharePoint — Sale Master list with sample data	sp_sale_master.png
SharePoint — User Role Reference list	sp_user_roles.png
Power BI — Page 1 Sale Overview KPI cards	pbi_overview.png
Power BI — Page 4 Readiness Score at 100%	pbi_readiness.png
Power BI — Data model view showing all table relationships	pbi_data_model.png
SQL — Script 02 output showing SLA labels	sql_sla_check.png

10. Future Enhancements

These items are out of scope for this portfolio build but represent realistic next steps for a production deployment.

- GUID-based IDs: Replace the CountRows SaleID formula with a Power Automate-generated GUID or sequential ID to eliminate the concurrency risk
- Dataverse backend: Migrate from SharePoint lists to Microsoft Dataverse for improved relational integrity, better delegation support, and row-level security
- Microsoft Teams integration: Replace email notifications with Teams Adaptive Cards for faster action and reduced inbox noise
- Power BI row-level security: Implement RLS so managers see only their direct reports' data
- Audit log automation: Replace the manual Change History list with a Power Automate flow that logs every status change automatically
- Employee self-service portal: Build a dedicated mobile-optimised screen with push notification support
- Multi-language support: Add a language variable and translation table in SharePoint to support international locations
- Capacity forecasting: Add a Power BI forecasting tab projecting participation volume based on historical request patterns